

Matthew G. Galarza

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EDUCATION

Rensselaer Polytechnic Institute

Doctoral candidate in Aerospace Engineering
Master of Engineering, Mechanical Engineering
Bachelor of Science, Mechanical Engineering

Dean's List 2018–2022, Rensselaer Recognition Award 2017–2022

Troy, NY
Expected Graduation: Dec. 2026
May 2022
May 2021

Relevant Coursework: Scientific Machine Learning, System Identification, System Analysis Techniques, Nonlinear Dynamics, Control Systems, Thermodynamics, Heat Transfer, Fluid Mechanics, Numerical Design Optimization, Finite Element Analysis.

WORK AND INTERNSHIP EXPERIENCE

GE Aerospace Research, *Mechatronics and Robotics Fellow*

Aug. – Dec. 2025

Developed biologically-inspired robotics-enabled inspection and metrology tooling for jet engine damage detection and dimensional verification, spanning Linipot-on-Demand (LOD) and Structured Light initiatives, reducing inspection time from ~50 hrs to ~2 hrs without teardown.

- Led concept-to-prototype development for Linipot-on-Demand (LOD), an in-situ probe for measuring stator distortion (core cold-clearance distortion) in assembled engines; drove multiple design evolutions via rapid prototyping and test-driven refinement.
- Designed compliant, soft-robotics-inspired mechanical interfaces to navigate tight borescope-port constraints and reduce risk during insertion; defined and validated an insertion protocol leveraging a borescope-port approach.
- Engineered the probe's anchoring and stabilization strategy to remain steady during rotation and enable repeatable measurements under real deployment constraints (engine geometry, articulation requirements; ~8 mm port diameter limiting instrumentation size).
- Advanced inspection workflow features including camera-based confirmation for visual detection of rubs/scabs/defects and faster deployment without teardown and reassembly (new/post-break-in/post-first-flight scenarios).
- Owned Structured Light data workflows: processed/cleaned multi-scanner datasets, performed point-cloud alignment/interpretation, and produced comparative trade studies (including a Pugh matrix) to drive scanner down-selection and procurement.
- Authored critical documentation to operationalize the work (SOP, user manual, assembly manual, BoM) and supported a live demo on an operational engine to senior managers and VPs at GE's Cincinnati, OH facility.
- Filed two patent applications / invention disclosures related to the inspection approaches [internal record IDs: 90522719, 90525767].

Collaborative Robotics, *Robotics Engineering Intern*

May – Dec. 2023

Played a pivotal role in supporting the successful launch of the company's inaugural pilot robot and securing \$30M series A funding round by spearheading the mechanical design, prototyping, and deployment of robotic systems.

- Led the design, manufacturing, and prototyping of a complete robotic subsystem crucial for perception and autonomy, aligning it with industrial design intent.
- Collaborated with manufacturers to provide valuable DFM/DFA feedback and streamline part procurement processes, optimizing efficiency and cost-effectiveness.
- Engineered and implemented testing rigs for system prototypes, ensuring a seamless transition from simulation to real-world operation, guaranteeing product reliability.
- Designed PCBs using Altium, providing essential support to hardware team bring-up.

Tesla Motors, *Mechanical/Process Engineering Intern*

May – Aug. 2021

Applied engineering fundamentals and a broad set of process engineering tools to solve technical problems and create detailed designs for a wide range of systems as well as providing engineering support to resolve process bottlenecks and other long-standing issues.

- Design and evaluation of pressure relief systems including pressure relief valves, emergency relief valves and pressure vacuum relief valves based on normal system safety operation limits and potential failure conditions.
- Performed cryogenic tank heat loss analysis to determine internal tank dimensions, highlighting factory vulnerability to delivery issues and potential production downtime; saved company over 45,000 dollars per month in preventative measures.
- Managed manufacturing and vendor relations to procure quotes, validate designs and facilitate device production.

Sanofi Pasteur, *Project Management Engineering Intern*

Jan. – Aug. 2020

Project team member within the Technical Services–Site Engineering department working on various GMP/Non-GMP projects to support vaccine manufacturing, maintenance, and facilities projects.

- Performed as-built field verification walk-downs and CAD revisions of over 100 engineering drawings including RO Water, WFI, AF&IDs, Formulation skids, piping, equipment, Autoclave, Clean Compressed, and Compressed air P&IDs.

- Supported a project diverting 5,122 tons of egg waste and 9,050 industrial plastic totes from landfills annually by assisting in the design and field testing of a rotating product chute for direct waste feed, troubleshooting issues, and recording data for analysis.

RESEARCH AND ACADEMIC EXPERIENCE

Nonlinear Dynamics and Data-Driven Optimization of Electromechanical Energy Harvesters Jan. 2024 – Dec. 2026

Advisors: Shaowu Pan, Yuri Lvov, and Diana-Andra Borca-Tasciuc

Graduate researcher working on the analysis and optimization of coupled nonlinear electromechanical energy harvesting systems, integrating applied mathematics and dynamical systems, control theory, and scientific machine learning techniques.

- Conducted in-depth analysis of coupled systems from an applied mathematics perspective, exploring energy conversion principles, nonlinear phenomena, mixing entropy, stability, and eigenvalue/eigenmode analysis using Lagrange, Hamiltonian, and Newtonian mechanics.
- Applied scientific machine learning techniques, including neural ordinary differential equations (neural ODEs) and universal differential equations (UDEs), to uncover latent dynamics and augment analytical models with data-driven corrections.
- Employed sparse system identification techniques (e.g., SINDy) to extract interpretable differential equations from experimental data, enabling targeted discovery of missing physics and refinement of hybrid gray-box models.

IBM Thin Films Fabrication Research

May 2024 – Dec. 2025

Advisor: Theodorian Borca-Tasciuc

Graduate research assistant sponsored by IBM to process and test semiconductor thin films in a cleanroom environment, contributing to research and development in next-generation semiconductor fabrication.

- Conducted wafer processing using photolithography, thickness measurements, e-beam evaporation, and liftoff to create samples for conductivity testing.
- Utilized 3ω , an electrothermal technique, to measure the thermal conductivity of processed thin film samples in both cross-plane and in-plane directions.
- Developed and implemented automated data processing scripts to streamline analysis and identify trends in material performance.

Electrostatic MEMS Energy Harvesting for Sensing Power

Jan. 2020 – May 2023

Advisor: Diana-Andra Borca-Tasciuc

Analytical and mathematical modeling for impact-based MEMS electrostatic frequency up-converter with topology optimization for enhanced energy harvesting and geometric scaling.

- Derivation of viscous fluid damping, electrostatic coupling, and collision dynamics to develop robust analytical models.
- Implemented Gaussian process Bayesian optimization through Latin Hypercube Sampling (LHS) to produce higher output voltages and provide insight into topological improvements.
- Fabricated devices in a cleanroom setting and tested device performance through frequency sweeps, applied g-force ranges, and DC voltage bias.

SELECTED PUBLICATIONS

A = ABSTRACT, C = CONFERENCE, J = JOURNAL, M = MANUSCRIPT IN PREPARATION

- [J.1] Jinglun Li, Habilou Ouro-Koura, Hannah Arnou, Arian Nowbahari, **Matthew Galarza**, Meg Obispo, Xing Tong, Mehdi Azadmehr, Einar Halvorsen, Mona M. Hella, John A. Tichy, and Diana-Andra Borca-Tasciuc, “Broadband Vibration-Based Energy Harvesting for Wireless Sensor Applications Using Frequency Upconversion,” *Sensors*, Vol. 23, No. 11, 5296 (2023). doi:10.3390/s23115296.
- [J.2] **Matthew G. Galarza**, Z. Sotoudeh, and D.-A. Borca-Tasciuc, “Investigation of Nonlinear Phenomena in Electrically Coupled Mechanical Oscillators With Applications in Electrostatic Energy Harvesting of Low Frequency Vibrations,” *Journal of Vibration and Acoustics*, Vol. 147, No. 5, 051006 (Oct. 2025). doi:10.1115/1.4068811.
- [C.1] Jinglun Li, Habilou Ouro-Koura, Hannah Arnou, Arian Nowbahari, **Matthew Galarza**, Meg Obispo, Xing Tong, Mehdi Azadmehr, Einar Halvorsen, Mona M. Hella, John A. Tichy, and Diana-Andra Borca-Tasciuc, “A Novel Comb Design for Enhanced Power and Bandwidth in Electrostatic MEMS Energy Converters,” in *Proc. IEEE 36th Int. Conf. on Micro-Electro Mechanical Systems (MEMS 2023)*, Munich, Germany, Jan. 15–19, 2023, pp. 728–731. doi:10.1109/MEMS49605.2023.10052590.
- [C.2] **Matthew G. Galarza** and D.-A. Borca-Tasciuc, “Investigation of Nonlinear Phenomena in Electrically Coupled Mechanical Oscillators With Applications in Electrostatic Energy Harvesting of Low Frequency Vibrations,” in *Proc. ASME Int. Mechanical Engineering Congress and Exposition (IMECE 2024)*, Portland, OR, USA, Nov. 17–21, 2024. Paper No. IMECE2024-88636.
- [C.3] Max Bloomfield, Amogh Wasti, Zongmin Yang, **Matthew Galarza**, Theodorian Borca-Tasciuc, Jacob Merson, Timothy Chainer, Prabudhya Roy Chowdhury, and Aakrati Jain, “A Multiscale Workflow for Thermal Analysis of 3DI Chip Stacks,” in *ITherm 2025 (24th Intersociety Conf. on Thermal and Thermomechanical Phenomena in Electronic Systems)*, Dallas, TX, USA, May 2025. doi:10.1109/ITherm55376.2025.11235683.
- [C.4] Amogh Wasti, Zongmin Yang, **Matthew Galarza**, Jonas Kendra, Davis Knight, Timothy Chainer, Roy Yu, Prabudhya Roy Chowdhury, Aakrati Jain, Max Bloomfield, Jacob Merson, and Theodorian Borca-Tasciuc, “Experimental Cross-Plane Thermal

Transport Characterization of BEOL Materials and Sensitivity to In-Plane Thermal Transport,” in *ITherm 2025*, Dallas, TX, USA, May 2025.

[C.5] Amogh Wasti, Zongmin Yang, **Matthew Galarza**, and Theodorian Borca-Tasciuc, “Effective Medium Approximation in Investigating Heat Transfer Through Multiple Sandwiched Layers of BEOL Materials,” in *Proc. ASME Heat Transfer Summer Conference (HT 2025)*, 2025. Paper No. HT2025-155734. doi:10.1115/HT2025-155734.

[C.6] K. Lodhiya, **Matthew G. Galarza**, Z. Sotoudeh, and D. Borca-Tasciuc, “Investigation of Energy Exchange Between Coupled Electro-Mechanical Conservative Oscillators,” in *Proc. ASME Aerospace Structures, Structural Dynamics, and Materials Conference (SSDM 2025)*, 2025. Paper No. SSDM2025-152235. doi:10.1115/SSDM2025-152235.

LEADERSHIP AND TEACHING EXPERIENCE

Graduate Teaching Assistant, RPI MANE Department

Jan. 2022 – Dec. 2025

- Held regularly scheduled office hours to advise and assist students.
- Assigned and evaluated student problem sets + exams, providing appropriate feedback when necessary.

Electronic Packaging Society (EPS) | IEEE, Member

Sept. – Dec. 2024

- Participated in introducing both undergraduate and early graduate students to the semiconductor industry.
- Held workshops, gave in-depth presentations on various topics and attended colloquiums held by industry professionals.

Rensselaer Motorsport | Formula SAE, RM26 Thermal System Design Lead

Sept. 2021 – May 2023

- Led a team of engineers to design, prototype and test various components systems on the RM vehicle.
- Collaborated across vehicle teams, placing a priority on the greater benefit of the vehicle above any one individual system.

TECHNICAL SKILLS

Programming Languages: Julia, Python (NumPy, Pandas, Seaborn, TensorFlow, PyTorch, Keras), MATLAB, Simulink, ANSYS ADPL, HTML, Mathematica

Software: Siemens NX, SolidWorks, PolyWorks, Altium, AutoCAD, Revit, Seeq, Maple, MS Office Suite Products

REFERENCES

Diana-Andra Borca-Tasciuc, Professor

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Shaowu Pan, Assistant Professor

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Jinglun Li, Senior Mechanical Engineer

Applied Materials
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Deepak Trivedi, Technology Manager

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Jack Erdozain, Platform Architecture Engineer

Apple

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